The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 June 1995

INCH-POUND

MIL-S-19500/371C
27 March 1995
SUPERSEDING
MIL-S-19500/371B
4 January 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER
TYPES 2N3902 AND 2N5157
JAN AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, high-power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-3).

1.3 Maximum ratings.

Type	P_T 1/ T_A = +25°C	P_T 2/ T_C = +75°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N3902	5.0	100	700	400	5.0	2.0	3.5	-65 to +200	1.25
2N5157	5.0	100	700	500	6.0	2.0	3.5	-65 to +200	1.25

1/ Derate linearly 29 mW/°C above T_A = +25°C.
2/ Derate linearly 0.8 W/°C above T_C = +75°C.

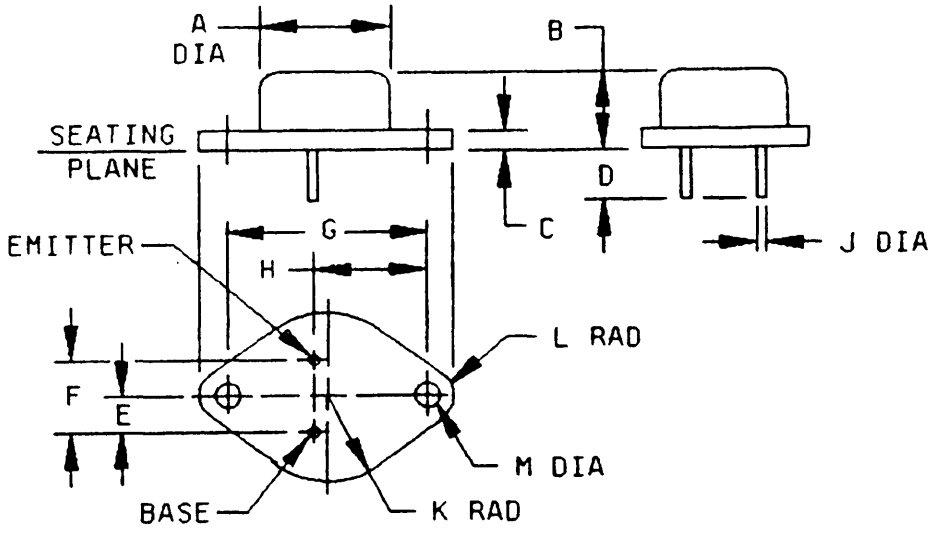
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A		.875		22.23	
B	.250	.328	6.35	8.33	
C	.060	.135	1.52	3.43	
D	.312	.500	7.92	12.70	3
E	.205	.225	5.21	5.72	
F	.420	.440	10.67	11.18	
G	1.177	1.197	29.90	30.40	
H	.655	.675	16.64	17.15	2
J	.038	.043	.97	1.09	3
K	.495	.525	12.57	13.34	
L	.131	.188	3.33	4.78	
M	.151	.161	3.84	4.09	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. This dimensions should be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement will be made at seating plane.
4. Two leads.
5. Collector shall be electrically connected to the case.

FIGURE 1. Physical dimensions.

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1.4 Primary electrical characteristics.

	h_{FE1} 1/	h_{FE2} 1/	$V_{BE(SAT)1}$	$V_{CE(SAT)1}$	C_{obo}	$\|h_{fe}\|$	Switching	
	V_{CE} = 5.0 V dc I_C = 0.5 A dc	V_{CE} = 5.0 V dc I_C = 1.0 A dc	I_C = 1.0 A dc I_B = 0.1 A dc	I_C = 1.0 A dc I_B = 0.1 A dc	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	V_{CE} = 10 V dc I_C = 0.2 A dc f = 1 MHz	t_{on}	t_{off}
			V dc	V dc	pF		µs	µs
Min	25	30				2.5		
Max		90	1.5	0.8	250	25	0.8	1.7

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX level). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX only
3	See 4.3.2
10	V_{CB} = 300 V
11	h_{FE2} and I_{CEX1}
12	See 4.3.1
13	ΔI_{CEX1} = 100 percent of initial value or 50 μA dc, whichever is greater. Δh_{FE2} = 25 percent of initial value; subgroup 2 of table I herein.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5; V_{CB} = 16-20 V dc; PT = 4.0 W.

4.3.2 Thermal response (ΔV_{BE} measurements). The ΔV_{BE} measurements shall be performed in accordance with MIL-STD-750, method 3131. The chosen ΔV_{BE} measurements and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen ΔV_{BE} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. One hundred percent safe operating area (SOA) testing may be performed in lieu of thermal response testing herein provided that appropriate condition of temperature, time, current, voltage, circuit and procedure to achieve die attach integrity are approved by the qualifying activity. The following parameter measurements shall apply:

a. I_M measurement - - - - - - - - - - - - - - - - - - 10 mA

b. V_{CE} measurement voltage - - - - - - - - - - - - - 20 V (same as V_H).

c. I_H collector heating current - - - - - - - - - - - 2 A (minimum).

d. V_H collector-emitter heating voltage - - - - - - - 20 V (minimum).

e. t_H heating time - - - - - - - - - - - - - - - - - - 10 ms (minimum).

f. t_{MD} measurement delay time - - - - - - - - - - - - 30 to 60 μs.

g. t_{SW} sample window time - - - - - - - - - - - - - - 10 μs (maximum).

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4 mA (minimum). c. I_H collector heating current

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with applicable steps in table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN and JANTX) of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
B3	1037	Bias condition (see 4.3.1), $t_{on} = t_{off} = 3$ minutes, 2,000 cycles.
B5	3131	$V_{CE} = 20$ V dc, $I_C = 2.0$ A, $R_{\theta JC} = 1.25°C/W$ maximum."

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
C2	2036	test condition A, weight = 10 pounds, t = 15 s.
C6	1037	Bias condition (see 4.3.1), $t_{on} = t_{off} = 3$ minutes, 6,000 cycles."

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 700 V dc; V_{BE} = 1.5 V dc	I_{CEX1}		500	µA dc
Emitter to base cutoff current	3061	Bias condition D;	I_{EBO1}			
2N3902		V_{EB} = 5.0 V dc			200	µA dc
2N5157		V_{EB} = 6.0 V dc			200	µA dc
Collector to emitter cutoff current	3041	Bias condition D;	I_{CEO}			
2N3902		V_{CE} = 325 V dc			250	µA dc
2N5157		V_{CE} = 400 V dc			250	µA dc
Base emitter voltage (saturated)	3066	Test condition A; I_C = 1.0 A dc; I_B = 0.1 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)1}$		1.5	V dc
Base emitter voltage (saturated)	3066	Test condition A; I_C = 3.5 A dc; I_B = 0.7 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)2}$		2.0	V dc
Collector to emitter saturated voltage	3071	I_C = 1.0 A dc; I_B = 0.1 A dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.8	V dc
Collector to emitter saturated voltage	3071	I_C = 3.5 A dc; I_B = 0.7 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		2.5	V dc
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 0.5 A dc; pulsed (see 4.5.1)	h_{FE1}	25		
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE2}	30	90	
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 2.5 A dc; pulsed (see 4.5.1)	h_{FE3}	10		
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 3.5 A dc; pulsed (see 4.5.1)	h_{FE4}	5		
Collector to emitter sustaining voltage		I_C = 100 mA dc	$V_{CEO(SUS)}$			
2N3902				325		V dc
2N5157				400		V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc	I_{CEX2}			
2N3902		V_{CE} = 400 V dc		300		µA dc
2N5157		V_{CE} = 500 V dc		300		µA dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE5}	10		
Subgroup 4						
Pulse response	3251	Test condition A except test circuit and pulse requirements in accordance with figure 2 herein.				
Turn-on time		V_{CC} = 125 V dc; I_C = 1.0 A dc; I_{B1} = 0.1 A dc	t_{on}		0.8	µs
Turn-off time		V_{CC} = 125 V dc; I_C = 1.0 A dc; I_{B1} = 0.1 A dc; $-I_{B2}$ = 0.50 A dc	t_{off}		1.7	µs
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 0.2 A dc; f = 1 MHz	$\|h_{fe}\|$	2.5	25	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		250	pF
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = 25°C; t ≥ 1 s; (see figure 3)				
Test 1		V_{CE} = 28.6 V dc; I_C = 3.5 A dc				
Test 2		V_{CE} = 70 V dc; I_C = 1.43 A dc				
Test 3						
2N3902		V_{CE} = 325 V dc; I_C = 55 mA dc				
2N5157		V_{CE} = 400 V dc; I_C = 35 mA dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 continued						
Safe operating area (switching)	3053	Load condition C (unclamped inductive load) (see figure 4) T_C = 25°C; duty cycle ≤ 10 percent; R_S = 0.1Ω (see 4.5.1)				
Test 1		t_p approximately 3 ms (vary to obtain I_C); R_{BB1} = 20Ω; V_{BB1} = 10 V dc; R_{BB2} = 3K ohms; V_{BB2} = 1.5 V dc; I_C = 3.5 A dc; V_{CC} = 50 V dc; L = 60 mH; R = 3 ohms; R_L ≤ 14 ohms.				
Test 2		t_p approximately 3 ms (vary to obtain I_C); R_{BB1} = 100Ω; V_{BB1} = 10 V dc; R_{BB2} = 3K ohms; V_{BB2} = 1.5 V dc; I_C = 0.6 A dc; V_{CC} = 50 V dc; L = 200 mH; R = 8 ohms; R_L ≤ 83 ohms.				
Safe operating area (switching)	3053	Clamped inductive load (see figure 5); T_C = +25°C; duty cycle ≤ 10 percent; t_p = approximately 30 ms (vary to obtain I_C); R_S = 0.1Ω; R_{BB1} = 20Ω; V_{BB1} = 10 V dc; R_{BB2} = 100 ohms; V_{BB2} = 1.5 V dc; V_{CC} = 50 V dc; I_C = 3.5 A dc; (see figure 5); R_L ≥ 0 ohms; L = 60 mH; R = 3 ohms A suitable clamping circuit or diode can be used. (see 4.5.1).				
2N3902		Clamp voltage = 400 +0, -5 V dc				
2N5157		Clamp voltage = 500 +0, -5 V dc				
		(Clamped voltage must be reached)				
Electrical measurements		See table II, steps 1, 3, and 4				

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Condition		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 700 V dc; V_{BE} = 1.5 V dc;	I_{CEX1}		500	µA dc
2.	Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 700 V dc; V_{BE} = 1.5 V dc	I_{CEX3}		400	µA dc
3.	Forward current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE2}	30	90	
4.	Emitter to base cutoff current 2N3902 2N5157	3061	Bias condition D; V_{EB} = 5.0 V dc V_{EB} = 6.0 V dc	I_{EBO1}		 200 200	 µA dc µA dc
5.	Emitter to base cutoff current 2N3902 2N5157	3061	Bias condition D; V_{EB} = 5.0 V dc V_{EB} = 6.0 V dc	I_{EBO2}		 400 400	 µA dc µA dc
6.	Forward current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	Δh_{FE1}	±25% change from previously measured value		
7.	Thermal response 3/	3131	See 4.3.2	V_{BE}			4/

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 3, and 4.

b. Subgroup 3, see table II herein, steps 2, 5, 6, and 7.

c. Subgroup 6, see table II herein, steps 2, 5, and 6.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 3, and 4.

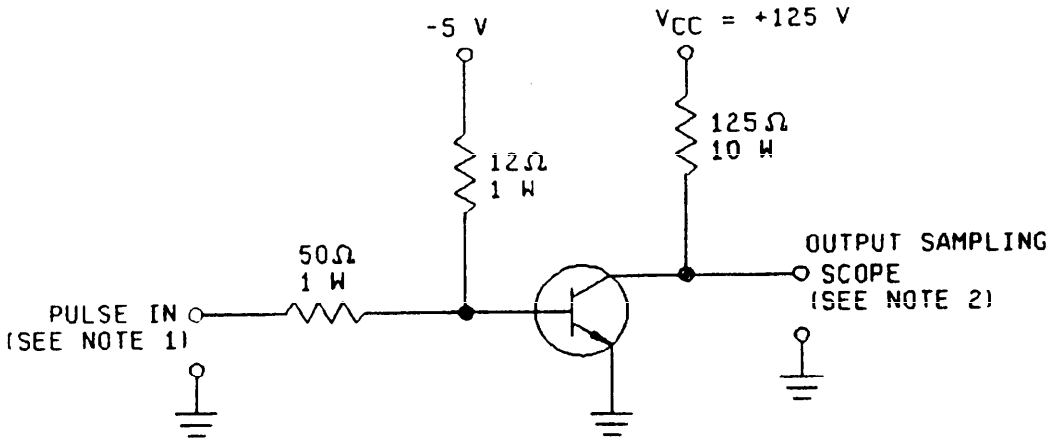
b. Subgroup 3, see table II herein, steps 1, 3, and 4.

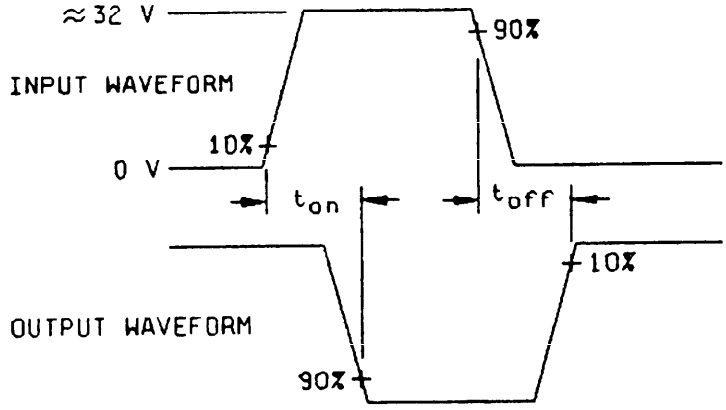
c. Subgroup 6, see table II herein, steps 2, 5, 6, and 7.

3/ Safe operating area (SOA) may be performed in lieu of thermal response testing herein provided that the appropriate conditions of temperature, time, current, voltage, procedure and circuit to achieve die attach integrity, are approved by the qualifying activity.

4/ Twenty percent degradation in group B is permitted; 30 percent degradation in group C is permitted.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 20 nanoseconds; duty cycle ≤ 5%; generator source impedance shall be 50Ω; pulse width = 5 μs.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 50 pF; rise time ≤ 2.0 ns.

FIGURE 2. Pulse response test circuit.

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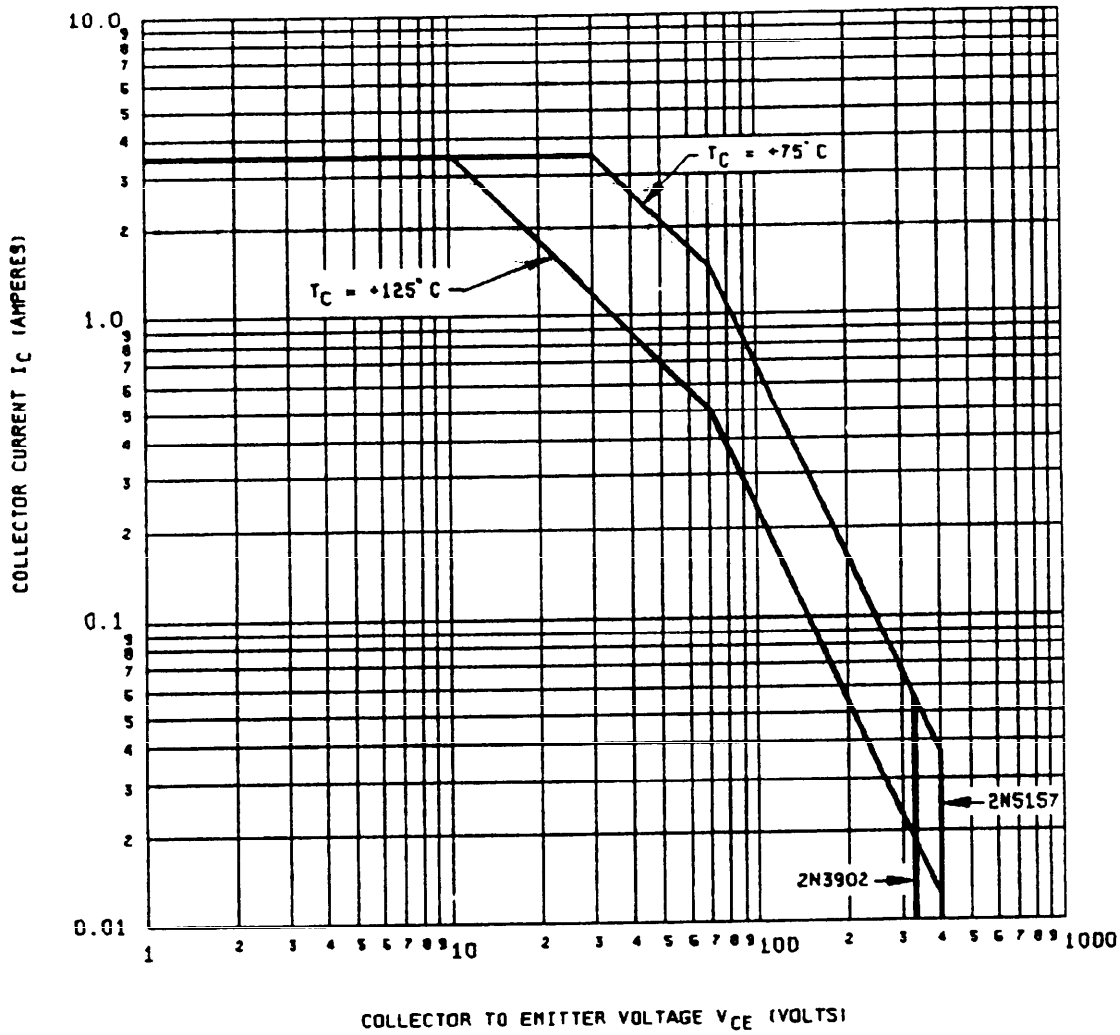


FIGURE 3. Maximum safe operating graph (continuous dc).

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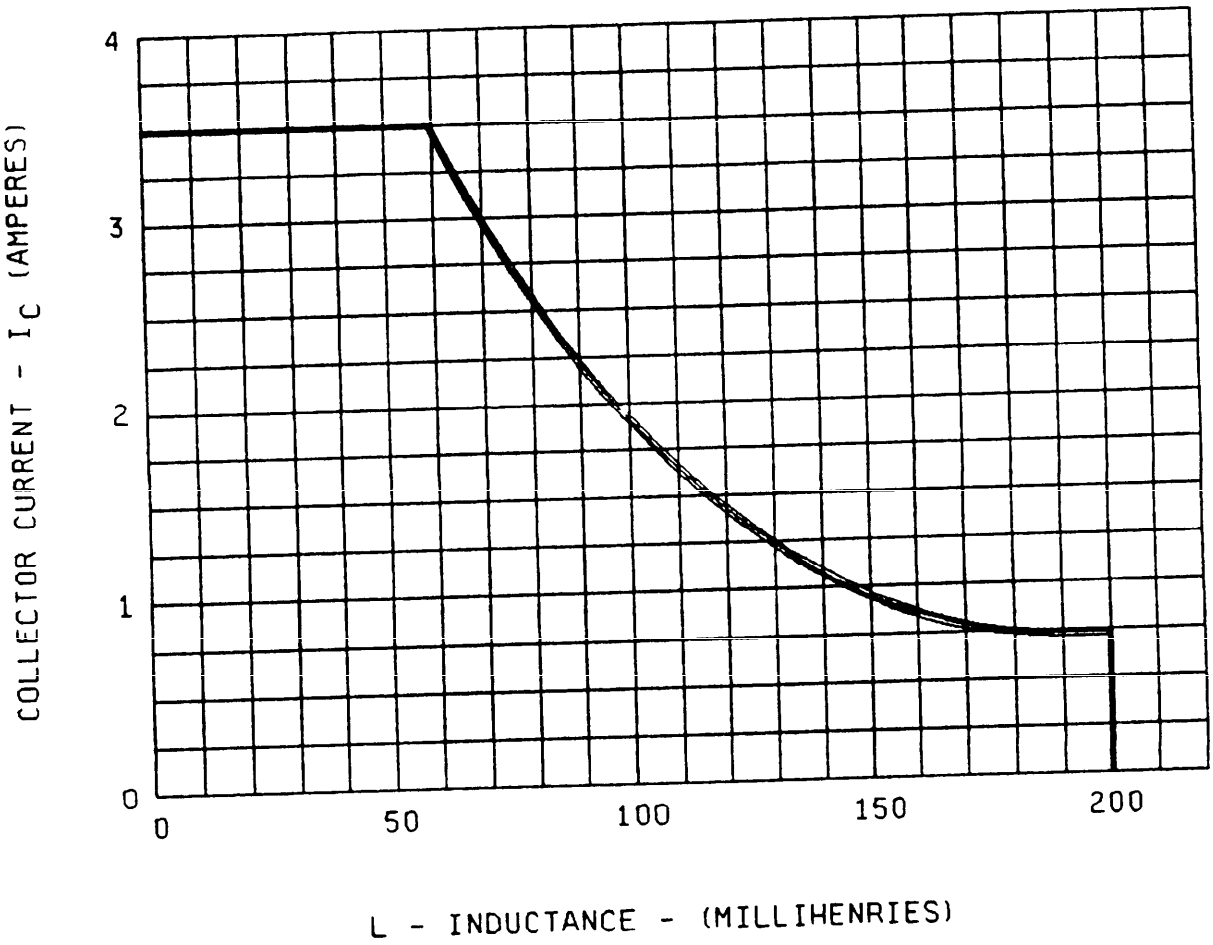


FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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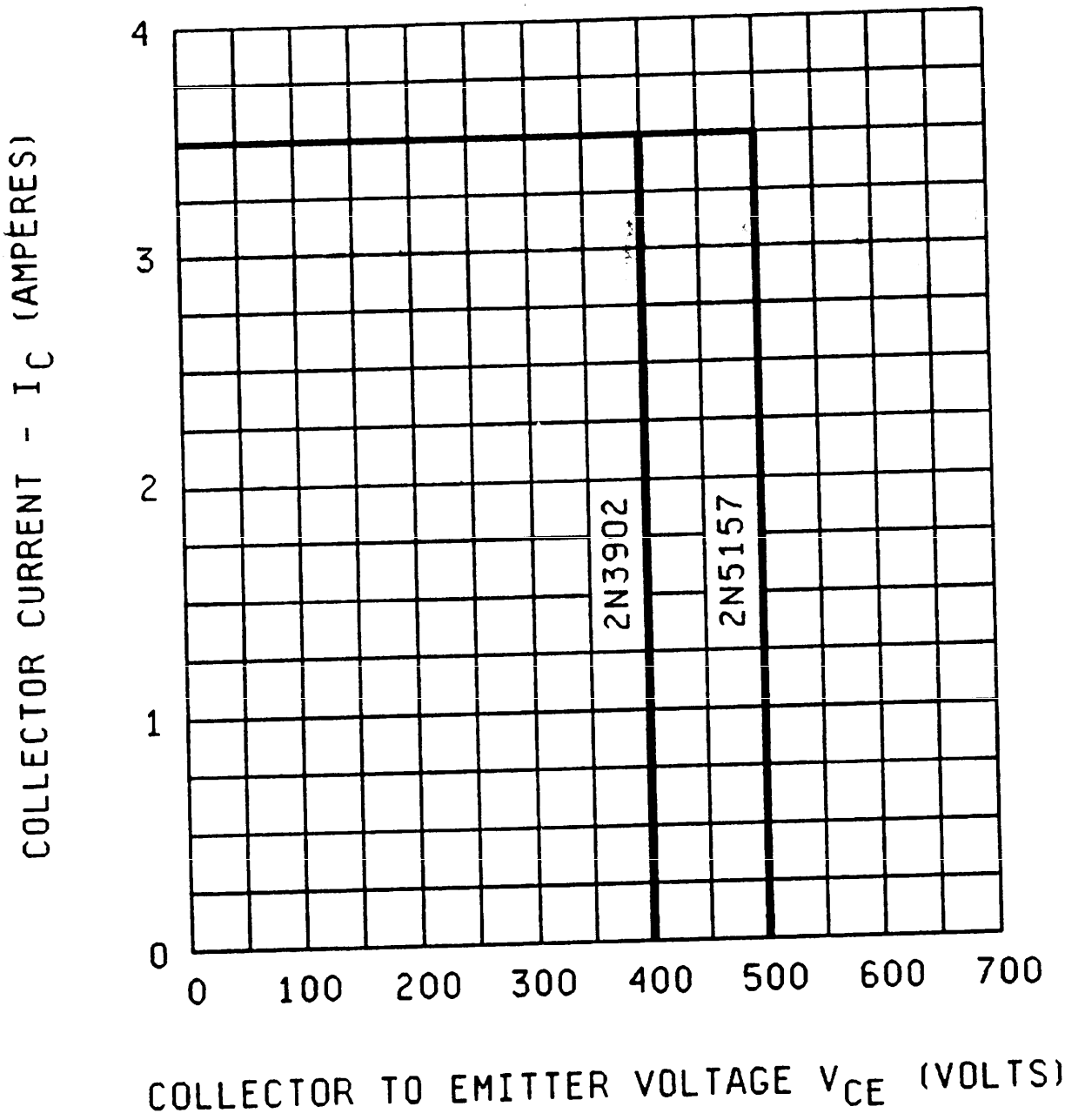


FIGURE 5. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Air Force - 19, 85, 99
Navy - AS, CG, MC

Preparing activity:
DLA - ES

(Project 5961-1629)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/371C	2. DOCUMENT DATE (YYMMDD) 95/03/27

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER TYPES 2N3902 AND 2N5157 JAN AND JANTX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048	(2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center ATTN: DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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